

A state government is planning a hydroelectric project on a river basin in addition to the production of electric power. The project will provide ~~the~~ flood control, irrigation and recreation benefits. The estimated benefit & cost are as follows:

initial cost = 8,00,00,000 Annual operation & maintenance cost = 30,00,000

Annual power sales = 60,00,000 Annual flood control saving = 30,00,000
Annual irrigation benefits = 50,00,000 Annual recreation benefits = 20,00,000
life of the project = 50 years

Check whether the project is justified by benefit cost ratio with $i = 12\%$.
Compounded annually.